

DHANVARDINI RAJENDRAN

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SUMMARY

Full stack data engineer with 3+ years building end to end data systems, from **Azure Databricks** and **Snowflake** pipelines processing billions of health plan records to **FastAPI** and **Django** services and **AI** features shipped to production. Strong across **ETL/ELT** design, orchestration, and data quality, plus the **API** and **AI** layers that turn pipelines into products.

WORK EXPERIENCE

Optum, UnitedHealth Group

July 2022 to Aug 2024

Software Engineer (Data Engineering and Analytics Platform)

- Built **10+ ETL pipelines** on **Azure Databricks** processing **4B+ pharmacy claims records**, publishing Gold layer **Delta** tables consumed by 5+ downstream teams.
- Reduced pipeline failure rate from **40% to 10%** through root cause analysis, **SQL** rewrites, and multi layered **data quality** gates.
- Developed **FastAPI** and **Django** services that exposed curated datasets and metrics to internal applications, replacing manual data pulls for teams supporting **5M+ health plan members**.
- Built **React** interfaces on top of those services, giving analysts self serve access to pipeline status and curated health plan data.
- Enhanced a multi step execution framework in **Python** and **PySpark** with reconciliation, deduplication, and rerun from failed step recovery.
- Deployed data assets across Azure Databricks workspaces through **Unity Catalog** with fine grained access control and cross workspace lineage, applying predicate pushdown and selective aggregation on billion row Delta datasets.

PROJECTS

PodcastIQ: Snowflake Native Agentic Search Platform

Mar 2026

- Architected a 4 layer **Snowflake** warehouse with **dbt Cloud** and **GitHub Actions CI/CD** powering an **agentic** semantic search agent with query routing, reaching 95.8% router accuracy at \$0.0012 per query.

IMDb Analytics: AWS Cloud Native Warehouse

Dec 2025

- Built an **AWS** cloud native ETL over **190M rows** from 7 datasets, landing S3 to **Glue** and modeling a **Redshift Serverless** star schema with 5 conformed dimensions, tuned with **DISTKEY** and **SORTKEY** and served through QuickSight.

DocuParse: SEC Document Intelligence Pipeline

Jul 2025

- Built a **DVC** orchestrated document intelligence pipeline for SEC filings using **Detectron2** and **LayoutLMv3**, extracting **187 tables** and **2,954 XBRL** concepts across 676 pages at \$1.05 per 1,000 pages.

META TradePulse: Real Time Trading Signal Platform

Apr 2025

- Engineered a **Kafka** and **PySpark** streaming pipeline computing technical indicators, persisted to **Snowflake** through Streams & Tasks for sub minute refresh.

EDUCATION

Northeastern University, College of Engineering

Aug 2024 to May 2026

Master of Science in Information Systems (GPA: 3.7/4.0)

Courses: Generative AI with LLMs in Data Engineering, Big Data Systems and Intelligent Analytics, Data Architecture for Business Intelligence, Data Management and Database Design

SASTRA University

Jun 2018 to Jun 2022

Bachelor of Technology in Computer Science

CERTIFICATIONS AND RECOGNITION

- AWS Certified Data Engineer** (Mar 2026), **AWS Certified Solutions Architect** (Jan 2026), **Microsoft Certified: Azure Data Fundamentals** (DP 900).
- Laurel and Scroll 100**: top 100 students across all Northeastern campuses for leadership and academic excellence.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, JavaScript, Java, FastAPI, Django, Node.js, Express, REST, Redis

Data Engineering & Streaming: PySpark, dbt, Azure Data Factory, Azure Databricks, Snowflake, Delta Lake, Kafka, Airflow, DVC, GitHub Actions CI/CD

AI: LLM and RAG pipelines, agentic workflows, Snowflake Cortex, LangChain, LlamaIndex, embeddings, Pinecone, Claude

Databases & Apps: SQL Server, PostgreSQL, MongoDB, Neo4j, React, Next.js, Streamlit, Plotly

Cloud & BI: AWS (S3, Glue, Redshift, QuickSight), GCP BigQuery, Power BI, Tableau